Lecture 27 — Matrix multiplication

Mon Oct 26

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Boyd video (optional): Lecture 27 — matrix multiplication · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 10.1. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Determine whether a product is defined and give its dimensions.
2. Compute a small matrix product.
3. Interpret ABx as a sequence of transformations.
4. Explain why matrix multiplication is generally not commutative.

Warm-up

Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}, \qquad x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

(a) What are the dimensions of A and of B? Is AB defined? If so, what size is it?

(b) Compute Bx (a 2-vector). You will need this later as "B first, then A."

Fill in

If A is $m \times n$ and B is $n \times p$, the product AB is the $m \times p$ matrix whose jth column is

$$(AB)_j = _____.$$

Equivalently, entry (i, j) is the inner product of _____ of A with _____ of B:

$$(AB)_{ij} = a_i^{\mathsf{T}} b_j.$$

When is AB defined? The inner dimensions must match: number of _____ of A equals number of _____ of B.

Associativity (for vectors). For a compatible p-vector x,

$$(AB)x = A(Bx).$$

Reading right to left: apply _____ first, then _____. So ABx is a **sequence of transformations**.

Not commutative. In general $AB \neq BA$ (even when both products are defined). Order of maps matters: "rotate then reflect" is not the same as "reflect then rotate."

Worked in class

W1. Dimensions. For each pair, decide whether AB is defined and give the size of AB if it is.

(a) A is 3×2, B is 2×4.

(b) A is 2×3, B is 2×4.

(c) A is 2×2, B is 2×2.

W2. Compute a product. With

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix},$$

compute AB by forming columns Ab_1 and Ab_2 (or by row–column inner products).

W3. Sequence of maps. Using the same A, B, and $x = (1, -1)$ from the warm-up, compute Bx, then $A(Bx)$. Compare with $(AB)x$ using your AB from W2. Same answer?

W4. Non-commutativity. With the same A and B, compute BA. Is $BA = AB$?

Your turn

Y1. Let

$$C = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & -1 \end{bmatrix}.$$

Is CD defined? Size? Compute CD.

Y2. Is DC defined? If so, give its size (you need not compute every entry unless time allows). What does this say about "CD vs DC"?

Y3. Let $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90° CCW) and $S = \mathrm{diag}(2, 1)$ (horizontal stretch). Compute RSx and SRx for $x = (1, 0)$. Same point? Interpret in one sentence.

Y4. True or false, with a one-line reason: if AB and BA are both defined, then A and B must be square of the same size.

Answers (Your turn)

Y1. C is 2×3, D is 3×2, so CD is defined and 2×2:

$$CD = \begin{bmatrix} -2 & 2 \\ 1 & 2 \end{bmatrix}.$$

(Columns: $Cd_1 = (-2, 1)$, $Cd_2 = (2, 2)$.)

Y2. Yes — D is 3×2 and C is 2×3, so DC is 3×3. Even when both products exist they need not have the same size, so they cannot be equal as matrices. Order matters.

Y3. $Sx = (2, 0)$, then $RSx = (0, 2)$. Meanwhile $Rx = (0, 1)$, then $SRx = (0, 1)$. Different: stretch-then-rotate $\neq$ rotate-then-stretch for this x.

Y4. **False.** Example: A is 2×3 and B is 3×2 — both AB (2×2) and BA (3×3) are defined, but neither factor is square. (They become the same size only when $m = n$.)